PRESENTER:
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BACKGROUND:

- IPV is a widespread but under-recognized public health issue
- 1 in 4 women and 1 in 10 men experience IPV in their lifetime.
- IPV includes physical, emotional, and sexual abuse
- IPV is associated with PTSD, depression, substance misuse, and suicide risk
- Traditional screening fails because many survivors underreport abuse due to stigma and fear
- Traditional ML models struggle with implicit emotional and sexual abuse detection
- LLMs are increasingly used in healthcare NLP for mental health, suicide risk, and clinical decision support

METHODS:

1. **Traditional ML -** RF, SVM, Logistic Regression, Naive Bayes -> Universal Sentence Encoder (USE) embeddings
2. **Baseline LLMs -** Llama-3.1-8B-Instruct, Qwen-2.5-7B-Instruct -> zero-shot, system prompt only, no training data used
3. **Prompting Strategies -** Zero-shot → Few-shot → Chain-of-Thought → RAG → Dynamic Few-shot → Advanced RAG
4. **LoRA Fine-Tuning -** Parameter-efficient fine-tuning on IPV corpus — only ~0.1% of parameters updated. Multi-task: trains all 4 subtypes simultaneously. Loss applied to output tokens only.
5. **Embedding Optimization -** Baseline: all-mpnet-base-v2 · Upgraded: Qwen3-Embedding-8B with supervised contrastive LoRA fine-tuning.
6. **Scale Comparison -** Llama-3.1-70B, Qwen-2.5-72B, GPT-5.2, GPT-5.4-pro -> same prompts and prompting strategies
7. **Evaluation Metrics -** Precision, Recall

QUALITATIVE EXAMPLES DETECTED ONLY AFTER FINE-TUNING:

Implicit Physical Coercion

"We were both yelling at each other, I tried to leave but she wouldn't let me pass.."

Subtle Emotional Manipulation

"Cheats on me with a guy at camp."

Detection of Intimate Partner Violence with Large Language Models

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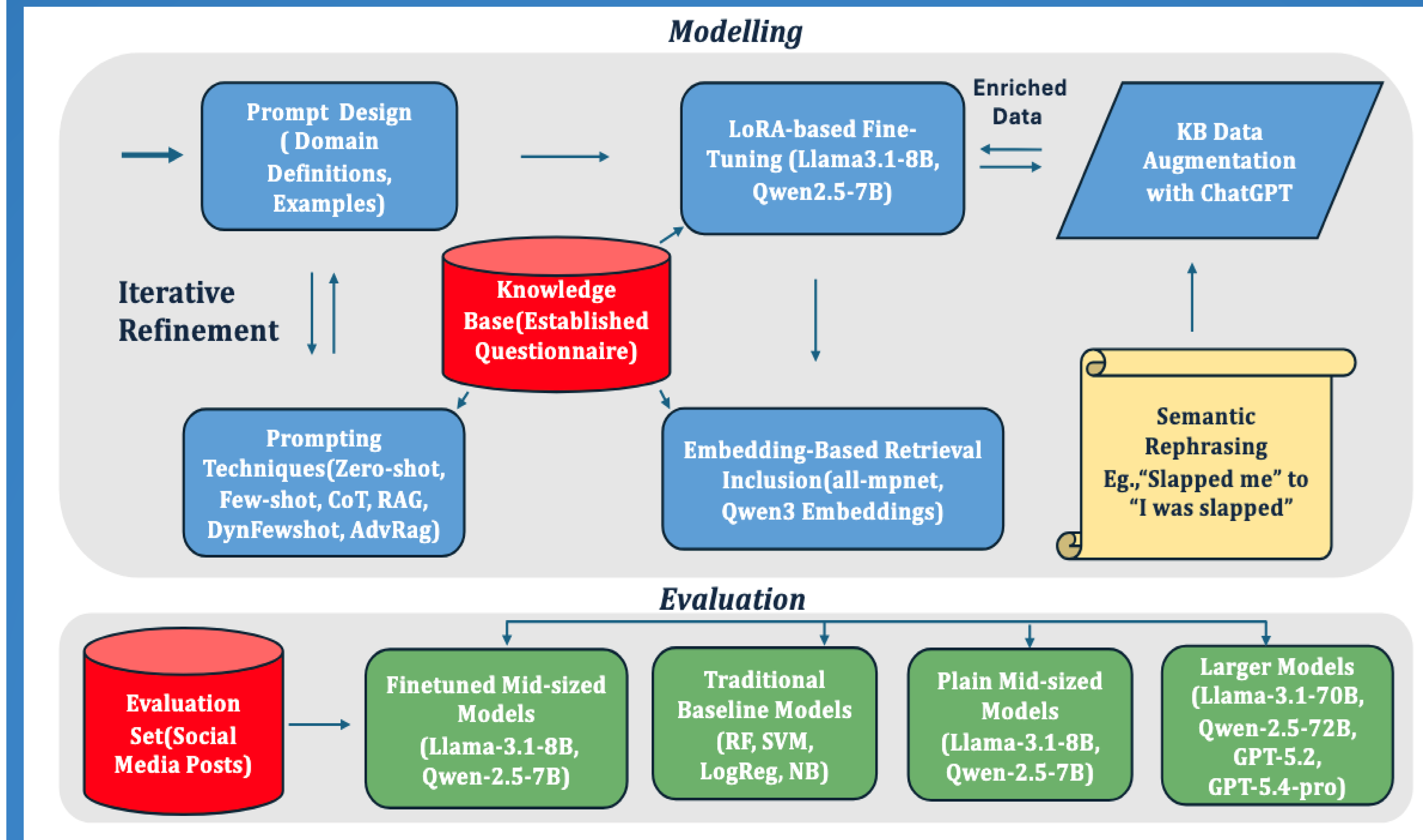
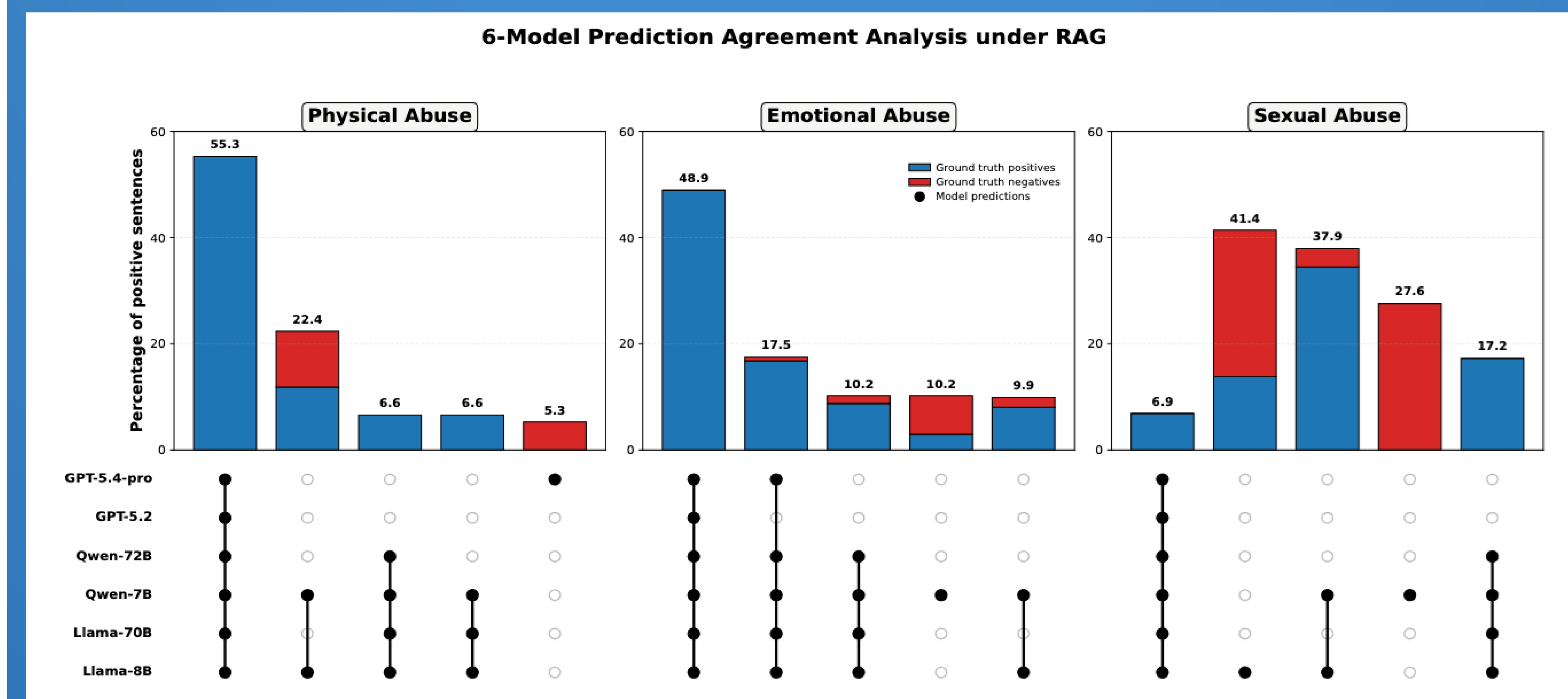


LLMs detect sexual and emotional abuse with near-perfect recall, bridging the gap where traditional machine learning and clinical tools fail.

Domain-adapted mid-sized LLMs outperform much larger frontier models for subtle IPV detection.



RESULTS:

- ML models are strong on physical abuse but catastrophic on sexual abuse
- Baseline LLMs already outperform traditional ML without task-specific training
- Retrieval-based prompting strategies provide the most consistent performance gains
- Larger frontier models improve precision but still fail on subtle abuse detection
- LoRA fine-tuning delivers the strongest and most stable performance improvements
- Better embedding models improve retrieval quality and prediction consistency
- **SEXUAL ABUSE RECALL:**

Traditional ML → 0.07

GPT-5.4-pro → 0.24

Llama-8B + LoRA → 1.00

Supports IPV screening, early intervention for family health practitioners, psycho-educational tool development, and teen abuse detection from social media platforms

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